Jewel G. Padilla

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Education

North Carolina State University, *Raleigh, NC* **Exp. May 2029**
Ph.D. in Human Factors and Applied Cognition
Advisor: Dr. Colleen E. Patton

North Carolina State University, *Raleigh, NC* **December 2024**
Bachelor of Arts in Psychology, *summa cum laude*
GPA: 4.0
Dean's List: Spring 2023, Fall 2023, Spring 2024

Research Interests

My research examines how people decide when to engage automation and artificial intelligence, and how those engagement decisions can be calibrated to support performance. I study discretionary automation use, perceived difficulty, trust and reliance, and automation bias through controlled laboratory experiments and applied evaluations of AI decision-support and sensemaking systems.

Keywords: human factors, human-automation interaction, human-AI teaming, discretionary automation, perception, trust and reliance, automation bias, expertise, risk, workload

Publications

Peer-Reviewed Conference Proceedings

- **Padilla, J. G.**, & Patton, C. E. (2025). Perceptions of difficulty in an automation-supported task. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 69*(1). https://doi.org/10.1177/10711813251360718
- **Padilla, J. G.**, & Patton, C. E. (in press). The impact of time pressure on perceptions of difficulty in an automation-supported task. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 70*(1).
- **Padilla, J. G.**, & Patton, C. E. (in press). Knowing when to use automation: Task-specific expertise and GPS engagement in a navigational task. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 70*(1).
- Perez-Cabarcas, M. M., Yakar, U. M., Hsu, I., Josyula, K. V., Lee, R. A., Harris, C., Destin, A. M., **Padilla, J. G.**, Addison, I. P., Mao, M., Lin, X., Mohsenin, T., & McDowell, K. (in press). COVERT-AI: A trustworthy multimodal system for adaptive real-time cognitive support. *Proceedings of the International Conference on Human-Computer Interaction (HCII 2026)*.

Technical Reports

- **Padilla, J. G.**, Barea, A., Johnson, B., & Tarosky, J. (2026). *makesense: A governed offline interface for multi-source intelligence sensemaking* [Technical report]. Summer Conference on Applied Data Science, Laboratory for Analytic Sciences, North Carolina State University.

Research Experience

Graduate Researcher **June 2025 – Current**
CHAT Lab, Human Factors and Applied Cognition Program, NC State University

- Design and conduct experiments on human-automation interaction and risk perception, including paradigm development, stimulus generation, and IRB protocol preparation and submission.
- Developing a second-year project comparing adaptable, adaptive, and fixed automation allocation authority in a within-subjects design.
- Apply parametric and non-parametric statistical tests (t-tests, ANOVA, regression, Wilcoxon) with assumption checking and effect-size estimation to evaluate behavioral patterns.
- Build analysis and visualization workflows in R to translate experimental data into structured metrics and present findings to diverse audiences.
- Authored and presented three first-authored papers at the Human Factors and Ergonomics Society (HFES) Annual Meeting.

Research and Data Science Intern **June 2026 – July 2026**
Summer Conference on Applied Data Science (SCADS), Laboratory for Analytic Sciences, NC State University

- First-authored the technical report on makesense, a governed, fully offline Model Context Protocol (MCP) server that exposes heterogeneous stores (relational, graph, search, document) to an agentic coding harness for multi-source intelligence sensemaking.
- Contributed to the makesense codebase by adding full-text PDF retrieval, resolving a disk-budget performance bottleneck with a regression guard, and extending continuous integration to run cross-platform on Windows.

- Designed and implemented a server-driven, three-choice human-in-the-loop approval flow that pauses sensitive agent-issued queries for explicit analyst sign-off, translating trust and reliance principles into a concrete interaction mechanism.
- Built a multi-model LLM evaluation harness on the RepLiQA benchmark and ran controlled A/B guardrail experiments (hardened-prompt and read-first variants) to quantify retrieval-augmented question-answering accuracy and factual grounding across models.
- Designed a within-subjects 2×2 factorial protocol (block randomization, AB/BA counterbalancing, matched parallel-form packets, and phase-level timing) to evaluate the agentic harness against manual multi-system workflows across efficiency, performance under complexity, and traceability.
- Built a multi-method measurement battery integrating validated instruments (Raw NASA-TLX, Trust in Automation, SUS), a seeded-error paradigm for automation bias, and blinded inter-rater rubrics, with timing pilots and pre-specified sample size, exclusion, and multiplicity rules.
- Presented design decisions and evaluation plans to program stakeholders and cross-disciplinary collaborators, translating their feedback into changes to the approval flow, session-opening interaction, and evaluation protocol.

UX Researcher & Human Factors Consultant **September 2025 – January 2026**
Designcorps & NC State University, Raleigh, NC

- Led an interdisciplinary team as part of a graduate human factors course capstone in collaboration with the Department of Design at NC State University.
- Conducted UX research using heuristic evaluation, task analysis, and experimental testing to evaluate prototype tactile walking surface indicators (TWSIs) designed for users with visual impairments.
- Presented findings and design recommendations to stakeholders.

Human Cognitive Enhancement Research Intern **June 2025 – August 2025**
DEVCOM Army Research Laboratory (ARL), Humans in Complex Systems (HCxS), Aberdeen Proving Ground, MD

- Worked cross-functionally with engineers, computer scientists, and psychologists on COVERT-AI, a multimodal retrieval-augmented generation (RAG) cognitive assistant that supports mental model formation for users navigating information-dense environments.
- Designed and conducted formative user evaluations of the human-AI interaction layer, measuring decision accuracy, response latency, and reliance on system confidence cues to inform changes to confidence signaling, information hierarchy, and intervention timing.
- Transformed raw interaction logs and experimental data into structured, analysis-ready datasets, treating override rates, error-correction behavior, and response-time distributions as evaluative signals communicated to developers as quantitative metrics.
- Translated human factors principles into actionable system requirements during the platform's transition from a monolithic desktop application to a containerized microservice architecture with a React front end.
- Communicated findings through visualizations, technical summaries, presentations, posters, and technical reports for a multidisciplinary team of ML engineers, systems architects, data engineers, and government stakeholders.

CHAT Lab Manager **January 2025 – May 2025**
Human Factors and Applied Cognition Program, NC State University, Raleigh, NC

- Analyzed study data in R and Excel and prepared reports of findings.
- Assisted in writing extended abstracts and documents for conference submission.
- Trained and mentored research assistants on research and experiment protocols.
- Interviewed and hired research assistants.

CHAT Lab Undergraduate Student Researcher **January 2024 – December 2024**
Human Factors and Applied Cognition Program, NC State University, Raleigh, NC

- Collaborated with Dr. Colleen Patton and graduate students; maintained weekly logs, led article discussions, and provided feedback on pilot studies.
- Supported project development by reviewing literature and gaining hands-on experience in independent research on human-automation interaction.

LEMON Lab Research Study Assistant **May 2024 – June 2024**
Czech Academy of Sciences, Prague, Czech Republic

- Collaborated with researchers studying phonetic acquisition by conducting a literature review on infant babbling.
- Learned and assisted with methodologies for measuring early speech, including EEG and acoustic analysis.
- Observed EEG administration sessions and supported data collection and discussions of experimental design.

ERPAS Undergraduate Student Researcher **August 2023 – December 2023**
School Psychology, NC State University, Raleigh, NC

- Reviewed scholarly articles and chapters on educational research, educational theory, social justice, and community-engaged scholarship.
- Developed materials for research and service projects.

Independent Research Projects

The impact of time pressure on perceptions of difficulty in an automation-supported task **August 2025 – April 2026**
CHAT Lab, Human Factors and Applied Cognition Program, NC State University

- Manipulated time pressure in a visual search task and found that participants underused a perfectly reliable aid despite the accuracy cost, showing that people's choices to engage automation aren't driven only by performance
- Analyzed accuracy, response times, subjective difficulty ratings, and automation use across timed and untimed conditions, chose between paired t-tests and Wilcoxon signed-rank tests based on normality and reported Cohen's d effect sizes in R.
- Published the results in a first-authored HFES 2026 Proceedings paper.
- Delivered a lecture on the findings to a multidisciplinary audience of researchers and practitioners at the HFES 2026 Annual Meeting.

Knowing when to use automation: Task-specific expertise and GPS engagement **August 2024 – December 2025**
CHAT Lab, Human Factors and Applied Cognition Program, NC State University

- Identified a gap in expertise research, which has focused on high-stakes domains like aviation and medicine, by examining how task-specific expertise shapes engagement with everyday consumer automation (GPS).
- Designed an experimental paradigm, including tasks, maps, and procedures, to test how task-specific expertise influences GPS engagement.
- Prepared IRB materials (consent, debrief, and exemption forms), led data collection, and ran nonparametric analyses in R (Wilcoxon rank-sum tests and Kendall's tau correlations).
- Published the results in a first-authored HFES 2026 Proceedings paper.
- Delivered a lecture on the findings to a multidisciplinary audience of researchers and practitioners at the HFES 2026 Annual Meeting.

Perceptions of Difficulty in an Automation-Supported Task **September 2024 – August 2025**
CHAT Lab, Human Factors and Applied Cognition Program, NC State University

- Manipulated visual clutter to vary difficulty in a visual search task and found that participants used automation more when the task was harder, though not enough to improve performance; motivating a follow-up study designed to separate perceived difficulty from objective difficulty.
- Analyzed accuracy, subjective difficulty ratings, and automation use in R, and published the findings as a first-authored paper in the HFES 2025 Proceedings.
- Presented the findings as a lecture to a multidisciplinary audience of researchers and practitioners at the HFES 2025 Annual Meeting in Chicago.

Perception of Risk: Examining the Influence on Automated Vehicle Usage **April 2024 – August 2024**
CHAT Lab, Human Factors and Applied Cognition Program, NC State University

- Investigated the relationship between situational risk and trust in automated vehicles.
- Created and administered a two-phase survey to assess participants' perceptions of risk and preference for automated versus manual driving.
- Analyzed the data with t-tests, ANOVA, and regression in R and presented the findings at two research conferences.

Fellowships and Awards

Graduate Research Assistantship; *North Carolina State University, Human Factors & Applied Cognition* **2025 – Current**

- Competitive assistantship awarded to select Ph.D. students in the program.

Inclusive Excellence Award ($500); *Human Factors and Ergonomics Society, Perception and Performance Technical Group* **2026**

- Awarded to support attendance at HFES ASPIRE 2026.

University Graduate Fellowship ($4,000); *North Carolina State University, Office of the Provost* **2025 – 2026**

- Awarded to incoming doctoral students in support of Ph.D. study in Psychology.

Provost Experience Program ($1,000); *North Carolina State University, Office of Undergraduate Research* **2024**

- Funding to compensate undergraduate contributions to ongoing research in a faculty laboratory.

Undergraduate Research Travel Award; *North Carolina State University, Office of Undergraduate Research* **Fall 2024**

- Funding to present undergraduate research at a conference.

Undergraduate Research Assistant Grant; *North Carolina State University, Office of Undergraduate Research* **Summer 2024**

- Competitive funding to support an independent research project for one semester.

Teaching

Instructor, *PSY 420: Cognitive Processes, North Carolina State University* **June 2026 – August 2026**

- Design and teach an undergraduate cognitive psychology course covering attention, memory, perception, and decision-making.
- Build the full curriculum, including lectures, assignments, and assessments aligned to course learning objectives.
- Lead class sessions and mentor students through office hours and individualized feedback.

- Evaluate student performance and assign grades across the term.

Presentations and Posters

- **Padilla, J. G.**, & Patton, C. E. (2026). *Who uses automation? Effects of spatial ability, confidence, and GPS use* [Poster, accepted]. American Psychological Association Annual Convention, Washington, DC.
- Santana, O., **Padilla, J. G.**, & Patton, C. E. (2026). *The effects of confidence and spatial knowledge on GPS usage* [Poster]. NC State Undergraduate Research & Creativity Symposium, Raleigh, NC.
- **Padilla, J. G.**, & Patton, C. E. (2025). *Perceptions of difficulty in an automation-supported task* [Lecture]. Human Factors and Ergonomics Society Annual Meeting, Chicago, IL.
- **Padilla, J. G.** (2025). *COVERT-AI for future C2 and Army use* [Poster]. Army Research Laboratory Summer Research Develop Program Symposium, Aberdeen Proving Ground, MD.
- Perez-Cabarcas, M., Destin, A., Harris, C., **Padilla, J. G.**, Lee, R. A., Hsu, I., Yakar, U., & Josyula, K. (2025). *COVERT-AI for mental model formation* [Talk]. Army Research Laboratory Humans in Complex Systems (HCxS) Symposium, Aberdeen Proving Ground, MD.
- **Padilla, J. G.**, & Patton, C. E. (2024). *Perception of risk: Examining the influence on automated vehicle usage* [Poster]. NC State Summer 2024 Undergraduate Research & Creativity Symposium, Raleigh, NC.
- **Padilla, J. G.**, & Patton, C. E. (2024). *Impact of risk perception on automated vehicle utilization in driving scenarios* [Poster]. National Research Conference at Penn, Philadelphia, PA.

Leadership and Service

Vice President, *Human Factors and Ergonomics Society (HFES), NC State Student Chapter* **May 2026 – Current**

- Support the chapter president in planning professional development events, including guest speakers and workshops.
- Coordinate chapter meetings and member communications to sustain engagement across the student body.
- Assist in recruiting and onboarding new members to grow chapter participation.
- Represent the chapter in collaboration with the Human Factors and Applied Cognition program and national HFES.

Student Reviewer, *Human Factors and Ergonomics Society (HFES) Annual Meeting* **February 2026 – Current**

- Evaluated submissions for the HFES Annual Meeting across General Sessions and Human-AI-Robot Teaming (HART).
- Provided structured feedback to authors on methodology, analysis, and clarity.

Research Mentor, *CHAT Lab, Human Factors and Applied Cognition Program, NC State University* **August 2025 – Current**

- Mentored an undergraduate researcher supporting a human-automation interaction project, guiding the development of an independent subproject.
- Advised on study design, analysis, and presentation, culminating in a poster at the NC State Undergraduate Research & Creativity Symposium.
- Wrote letters of recommendation supporting the student's academic and professional applications.

Professional Memberships

- Student Member, Human Factors and Ergonomics Society (HFES)
- Student Member, American Psychological Association (APA)

Certifications

- CITI Program, Responsible Conduct of Research (2026)
- CITI Program, Human Subjects Research (2024)

Skills

- **Data Analysis & Statistics:** mixed-effects/multilevel models, ANOVA, regression, signal detection analysis, non-parametric tests (Wilcoxon, Mann-Whitney), effect-size estimation, power analysis
- **Experimental & Research Methods:** experimental paradigm design, stimulus generation, IRB protocol preparation and submission, pilot and engineering testing, NASA-TLX and workload measurement, survey design, literature review
- **User Research & Usability Testing:** usability testing, user interviews and surveys, heuristic evaluation, task analysis, cognitive walkthroughs
- **Domain Knowledge (Human Factors):** human-automation interaction, trust and reliance, automation bias, situation awareness and workload, human-AI teaming, perception, performance
- **Domain Knowledge (AI/ML Systems):** agentic AI and LLM orchestration, retrieval-augmented generation (RAG), Model Context Protocol (MCP) tool architecture, LLM evaluation and guardrails (A/B testing, control arms, cross-judge agreement), performance profiling, benchmark authoring
- **Programming & Tools:** R (RStudio), Python, SAS, SPSS, SQL/DuckDB, PsychoPy, Qualtrics, Git/GitHub (PR review, CI/CD), pytest, Claude Agent SDK, LaTeX, Unity, Microsoft Office Suite, Slack, Canva